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Paleomagnetism of Mesozoic Rocks From the Central Andes of Southern Peru: Importance of Rotations in the Development of the Bolivian Orocline

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A paleomagnetic study has been undertaken in Jurassic volcanics and Cretaceous intrusions from coastal southern Peru. A secondary magnetization in the Jurassic volcanics was identified at 21 sites from three main localities along the southern coast of Peru (mean virtual geomagnetic pole: 59.5°N, 190.1°E, k=111, α95=3.0). This secondary magnetization was acquired during a tectonomagmatic event that may be related to the emplacement of the Peruvian coastal batholith and/or to the first Andean phase of compressional deformation in Late Cretaceous. If we assume that the remagnetization is of Late Cretaceous age, a regional counterclockwise rotation of about 20° is recorded by the forearc of southern Peru from 16°S to 18°S. Primary remanent magnetizations recorded by the volcanics and the batholith reveal a more complex tectonic history of block rotations. Counterclockwise rotations of 50° to 80° are documented near the localities of Chala (eight sites), Oconà (two sites) and Arequipa (four sites). Preliminary results based on three sites of Late Permian to Early Triassic sediments from northern Bolivia also show a large counterclockwise rotation. The observed differences between the primary and the secondary magnetization indicate that large block rotations occurred prior to the acquisition of the secondary magnetization. Perhaps, this first episode of deformation partially structured the Andean chain in the Central Andes and induced an incipient orocline. During the Cenozoic, a differential shortening along the chain gave the present-day shape of the Bolivian deflection and resulted in a counterclockwise rotation of the forearc system by about 20° as indicated by the remagnetization.

INTRODUCTION

The Andes are considered a typical example of a mountain belt related to the subduction of an oceanic lithosphere beneath a continental plate [*Dewey and Bird*, 1970]. While subduction has been a continuous process since at least the Early Cretaceous, compressional deformations appear to have occurred during short-lived events that can be related to variations in relative convergences between the oceanic plate and the South American continent [*Pardo-Casas and Molnar*, 1987; *Soler and Bonhomme*, 1990] or to collision of allochthonous terranes well documented in the Northern Andes [*Feininger*, 1987]. In the Central Andes (from 10°S to 30°S), however, the origin of the wide and thick Central Andean Plateau and the strong curvature of the chain (i.e., the Bolivian orocline) are not well understood. Although magmatic addition of material from the mantle [*Thorpe et al.*, 1981] could explain the Andean crustal thickening, the hypothesis concerning important crustal shortenings in the Central Andes has been recently emphasized in an analysis of the Central Andean uplift [*Isacks*, 1988]. This hypothesis is supported by geological evidence in Peru [*Dalmayrac et al.*, 1980; *Mégard*, 1984]. Interpretations of gravity anomalies over the Bolivian Andes [*Lyon-Caen et al.*, 1985] and cross sections of the sub-Andes of Bolivia based on geophysical and geological data also indicate about 200 km of crustal shortening [*Roeder*, 1988; *Baby et al.*, 1989; *Sheffels*, 1990].

The idea that crustal shortening is associated with collision processes as in the Himalayan belt and the recognition of several accreted or displaced terranes along the western margin of North America [*Beck et al.*, 1980] have triggered new interest in paleomagnetic studies of the Andes. Geological evidence for allochthonous terranes in northern Peru and western coastal Ecuador has recently been confirmed by paleomagnetic studies [*Mourier et al.*, 1988; *Roperch et al.*, 1987]. Although several authors have suggested the possibility of displaced terranes in the Central Andes [*Howell et al.*, 1985], there is no clear evidence such as ophiolite rocks indicating that collision and accretion of exotic terranes have occurred during the Phanerozoic. The paleomagnetic studies undertaken in the Central Andes do not support the hypothesis of allochthonous terranes [*Palmer et al.*, 1980*a*, *b*; *Heki et al.*, 1984; *May and Butler*, 1985]. However, these studies have shown some angular discrepancies between Peru and Chile that led *Heki et al.* [1983*a*] and *Kono et al.* [1985] to propose a large oroclinal bending of the Andes. A differential amount of shortening with a maximum in the central Andes of 250 to 425 km also implies relative forearc rotation of 15° to 25° between Peru and Chile [*Isacks*, 1988]. Another interpretation of paleomagnetic rotations in Peru and Chile has been proposed by *Beck* [1988], who suggests that in situ block rotations result from oblique directions of convergence between the oceanic plate and the overriding continental crust. In this model, rotations are mostly confined to the fore-arc system. However, there is no important shear fault system well documented in the Andes of Peru, and the left-lateral displacement of Quaternary age documented by *Armijo and Thiele* [1990] along the Atacama fault zone in northern Chile is in contradiction with the right-lateral component predicted by models of the Nazca-South America present-day plate

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kinematics. Moreover, the obliquity of convergence between the Nazca plate and Peru during early Cenozoic time was different from the present day plate configuration [*Pardo-Casas and Molnar*, 1987].

To better understand the relative contribution of collision, shortening, and shear to tectonic rotation in the Central Andes, we still need a better spatial and temporal distribution of the paleomagnetic data. The Central Andes have a complex tectonic history, and various block rotations should be expected within domains bounded by thrust and strike-slip faults.

In order to constrain the hypothesis of a strong shortening in the Central Andes and the bending of the orocline, we have undertaken a paleomagnetic study of Jurassic volcanism and Cretaceous batholith outcropping along the coast of southern Peru. The coast range of southern Peru could have recorded the most significant rotations because (1) the western external units of the Andean chain should integrate all the displacement due to the shortening on the eastern side and (2) the strike of the Andean structures in southern Peru are the most deflected from a general N-S orientation. On the other hand, we decided to focus on the Mesozoic units because these rocks have been affected by all the Andean deformations. Results from Late Paleozoic to Early Mesozoic sediments near Lake Titicaca will also be discussed.

GEOLOGICAL SETTING

Results of structural and sedimentological studies in southern Peru and Bolivia [*Vicente*, 1981, 1989; *Vicente et al.*, 1979; *Jaillard et al.*, 1990; *Sempéré et al*; 1991] define several paleogeographic domains bounded by major structures which are, from west to east, the West Peruvian domain, the Altiplano, the eastern Cordillera and the eastern basins. In southern Peru, the West Peruvian domain is divided by the Early to Late Cretaceous Cincha-Lluta overthrust [*Vicente et al.*, 1979] which delimits the coastal range and the western Cordillera (Figure 1). The pre-Mesozoic basement of the coastal range consists of the Precambrian metamorphic complex of the Arequipa massif [*Shackleton et al.*, 1979] with a locally preserved Devonian to Permian cover, and it is restricted to the area west of the Cincha-Lluta overthrust.

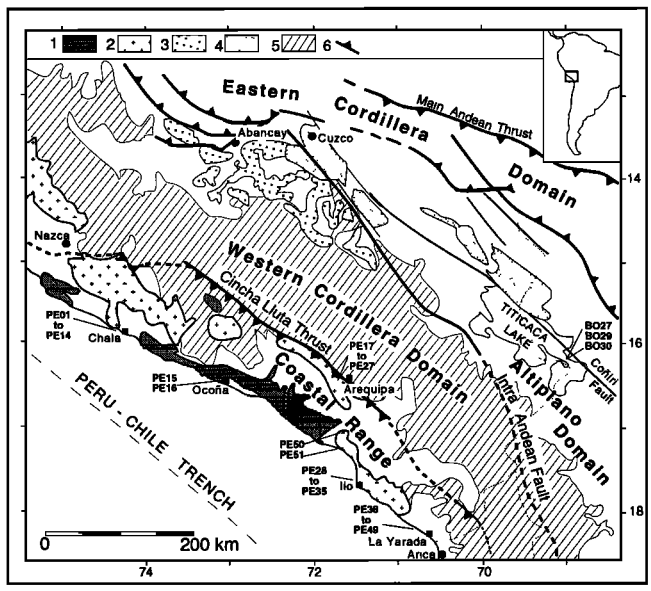


Fig. 1. Schematic geological map of southern Peru (modified from *Soler et al.* [1989]) 1, Precambrian basement; 2, Coastal Batholith; 3, Andahuaylas-Yauri batholith; 4, Red Beds continental basins; 5, Cenozoic volcanism; and 6, thrust fault.

The Mesozoic volcanics from the coastal range outcrop mostly along the coast. The paleomagnetic sampling in Peru was made at several distinct localities referred hereafter in the text and figures as Chala, Ocoña, Arequipa, Ilo, and La Yarada (Figure 1). Three sites in Late Permian and Early Triassic sediments [*Martinez*, 1980] from the Tiquina formation were also sampled near the lake Titicaca in Bolivia.

Mesozoic Volcanics

In southern Peru, the Mesozoic sequence is formed by several formations; the oldest is the Chocolate Formation which is overlain by the Guaneros Formation at Ilo and La Yarada. To the north at Chala and southwest of Nazca, two other formations (the Chala and the Rio Grande formations) are identified with distinctive characteristics from the Chocolate and Guaneros formations.

Near Ilo, the lower part of the Chocolate Formation is unconformably deposited on the Precambrian basement. In the Arequipa area, the upper part of the Chocolate Formation is characterized by andesitic volcanic breccias with interbedded marine and continental sedimentary sequences. In this area, an early Sinemurian faunal association is recorded at the top of the formation [*Vargas*, 1970; *Vicente*, 1981]. Near Ilo, the Chocolate Formation is intruded by the Jurassic gabbro-diorite rocks dated at 190 Ma and 150 Ma [*McBride*, 1977; *Sánchez*, 1983; *Mukasa*, 1986; *Clark et al.*, 1990]. Near la Yarada, volcanic breccias with intercalated ignimbritic flows of the Chocolate Formation are unconformably overlain by the Guaneros Formation [*Jaén et al.*, 1963]. This Bajocian to Callovian shallow marine series with basal basaltic flow constitutes the northern extension of the Camaraca Formation [*Salas et al.*, 1966]. An age of 130 Ma has been obtained by the $^{39-40}Ar$ method on a basaltic flow of the Guaneros Formation (PE44, Table 1). This age probably represents a partial reset by the Cretaceous plutonism of the area [*Clark et al.*, 1990]. In the Chala area, a thick sequence (>1500 m) is represented by basaltic to ignimbritic flows and dacitic breccias with intercalated red beds. These rocks were previously assigned to the Chala Formation and correlated with the Chocolate Formation [*Olchauski*, 1980]. Nevertheless, $^{39-40}Ar$ ages of 177 Ma and 157 Ma from basaltic flows (PE02 and PE08, Figure 2) indicate that the Chala Formation was deposited during the Middle Jurassic and might represent the southern extension of the Rio Grande Formation [*Ruegg*, 1956]. Various generations of andesitic sills and dikes crosscut the Jurassic formations of the Coastal Range. An $^{39-40}Ar$ age determination performed on a dike from the Ilo area (PE32, Figure 2) indicates magmatic activity contemporaneous with the Middle Jurassic sedimentation. Similar results have been obtained in the Rio Grande area [*Aguirre and Offler*, 1985; *Aguirre*, 1988]. Nevertheless, we cannot reject the hypothesis that some of these subvolcanic intrusions can represent the feeding channels of the volcanic activity of the Albian Casma Group [*Myers*, 1974; *Guevara*, 1980] or Paleogene Toquepala Formation [*Laughlin et al.*, 1968; *Bellon and Lefrèvre*, 1976]. Sediments and interbedded volcanics from the Chala and Guaneros formations are gently folded (15° to 40°), and the Chocolate Formation has a more steeply dipping attitude at Ilo

TABLE 1. K-Ar and $^{39-40}Ar$ Ages

	K (%)	radiogenic ^{40}Ar (%)	radiogenic ^{40}Ar ($\xi 10^{-5}$ $\chi\chi/\gamma$)	Age $\pm 1\sigma$ (Ma)
K-Ar				
PE16	2.523	91.6	1.6198	158.1±1.6
PE34	0.472	66.2	0.2335	123.0±1.3
$^{39-40}Ar$				
PE02 (integrated total age)				182.6
plateau age (steps 4-6)				177.1±2.2
PE08 (integrated total age)				153.6
plateau age (steps 4-6)				157.3±1.4
PE44 (integrated total age)				124.3
plateau age (steps 2-5)				129.9±0.3
PE32 (integrated total age)				159.7
plateau age (steps 2-5)				162.5±0.3

All radiometric datings were performed at Oregon State University by R. A. Duncan (J-factor = 0.00157).

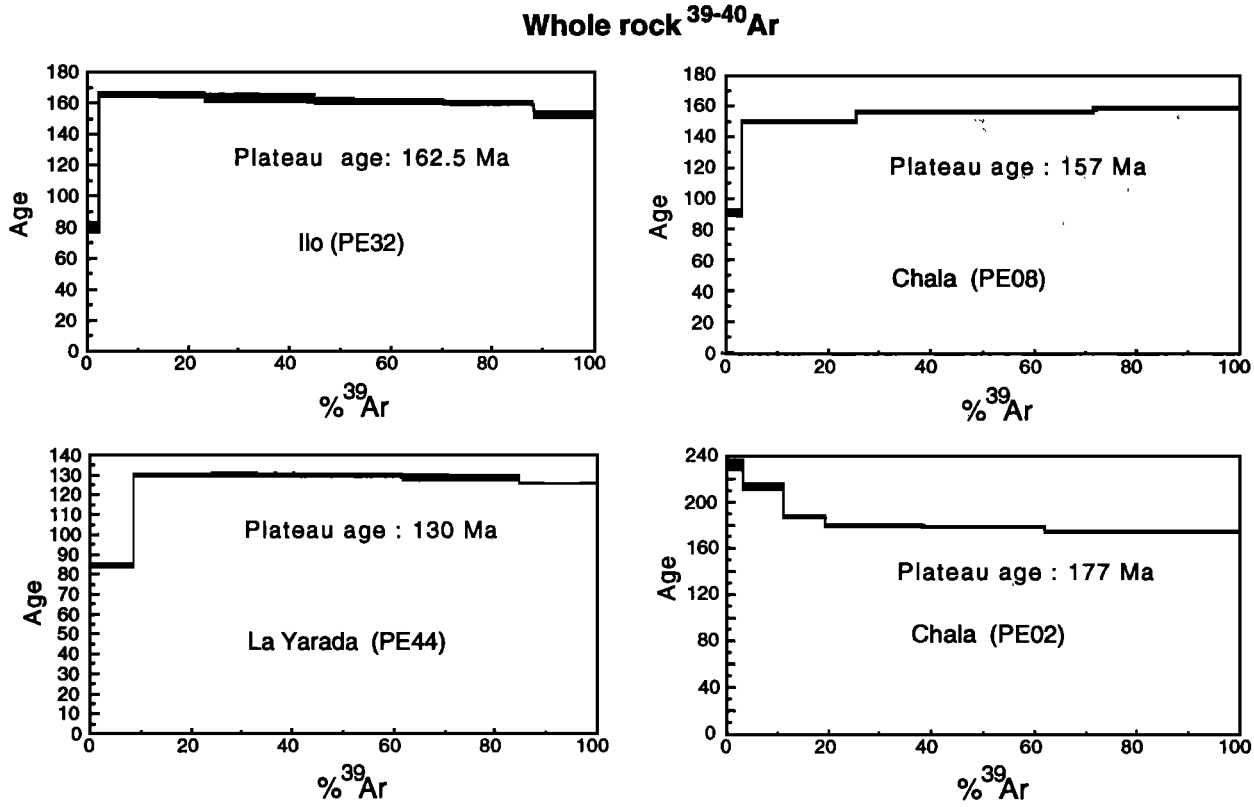


Fig. 2. Whole rock $^{39-40}Ar$ dating.

and la Yarada. The age of this deformation is unknown. Structural corrections are given for each site in Table 2.

Plutonic Rock

In Southern Peru, Triassic, Jurassic, and Cretaceous intrusions are confined to the West Peruvian Domain. At La Planchada, near Ocoña, where the Jurassic sequence was not deposited or was not preserved, localized monzonitic plutons intrude the Precambrian basement. We have obtained an age of 158 Ma by K-Ar on whole rock (PE16, Table 1) but a K-Ar age of 220 Ma on biotite has been recently obtained (Bonhomme, et al., unpublished data, *1991*); this difference in age suggests that these rocks have undergone partial resetting. At least three distinct plutonic episodes of Middle Jurassic (circa 185 Ma), Late Jurassic (circa 160-165 Ma) and Middle Cretaceous (circa 95-110 Ma) are reported in the Ilo-La Yarada area [*McBride*, 1977; *Sánchez*, 1983; *Mukasa*, 1986; *Clark et al.*, 1990]. An age of 123 Ma has been obtained by K-Ar on whole rock on a pluton near Ilo (PE34, Table 1). This age may represent a Jurassic intrusion partially reset by the Cretaceous plutonism of the area. In the Arequipa area, Jurassic and Cretaceous plutons are known [*Mukasa*, 1986], but the paleomagnetic sampling was restricted to the Cretaceous intrusions; the Early Cretaceous phases of the magmatic activity are dated at 105 Ma and the last ones at 61 Ma [*Mukasa*, 1986]. Four superunits are distinguished: the Early Gabbros (105-101 Ma), the Tiabaya superunit (84-78 Ma), the Linga-Yarabamba super-unit (71-62 Ma) and the Cerro Verde Quartz Monzonite (61 Ma) [*Stewart et*

TABLE 2. Site Mean Paleomagnetic Results

Site	Rock Type	Site Coordinates Longitude(W);Latitude(S)	χ	J_{NRM}	Secondary IS					Primary IS					Tilt Correction		Primary TC	
					N	*D*	*I*	*k*	α_{95}	*N*	*D*	*I*	*k*	α_{95}	Strike	Dip	*D*	*I*
								Chala										
PE01	basaltic flow	74°14'48";15°51'33"	0.1	0.005	4	325.9	-17.5	40	14.7	5	142.7	40.8	18	18.6	281	35	104.8	55.5
PE02	basaltic flow	74°14'48";15°51'30"	0.04	0.006	6	316.7	-18.8	99	6.8						281	35		
PE03	basaltic flow	74°14'48";15°51'36"	0.04	0.013	7	320.0	-29.1	49	8.7						281	35		
PE04	basaltic flow	74°13'36";15°52'05"	0.05	0.038	7	329.0	-24.0	151	4.9	7	294.7	-9.7	166	4.7	278	40	284.6	-18.1
PE05	andesitic dike	74°13'33";15°52'10"	5.7	0.540						5	113.5	17.2	277	4.6	272	45	94.5	27.2
PE06	basaltic flow	74°13'43";15°51'59"	0.06	0.009	6	331.8	-28.1	142	5.6	5	125.3	-2.2	487	3.5	273	32	122.3	14.5
PE07	basaltic flow	74°13'33";15°51'50"	0.05	0.013	7	324.2	-34.3	253	3.8	5	152.8	-43.7	356	4.1	273	32	160.9	-14.8
PE08	basaltic flow	74°12'23";15°51'28"	0.3	0.140						7	296.0	-18.2	79	6.8	258	30	283.1	-34.3
PE09	andesitic flow	74°09'48";15°51'32"	1.4	0.260	8	323.6	-22.9	235	3.6						247	27		
PE10	andesitic flow	74°09'48";15°51'24"	1.0	0.057	7	326.9	-21.6	248	3.8	3	140.1	40.1	483	5.6	247	27	125.3	65.0
PE11	ignimbritic flow	74°11'10";15°51'29"		0.120						5	118.2	21.3	68	9.4	283	32	103.9	25.9
PE12	red sandstone	74°14'48";15°51'17"	0.04	0.016	9	326.0	-29.4	392	2.6						239	16		
PE14	granodiorite	74°25'50";15°45'43"	3.1	0.260						8	128.6	7.2	56	7.5				
								Ocona										
PE15	monzodiorite	73°15'30";16°23'00"	0.05	0.019						5	96.1	52.5	45	11.6				
PE16	monzodiorite	73°12'50";16°23'25"	0.01	0.035						7	86.4	43.9	234	4.0				
								Arequipa										
PE17	gabbro	71°42'48";16°28'28"	2.7	0.160						8	310.4	-38.2	15	14.8				
PE19	gabbro	71°42'48";16°28'56"	2.3	0.250						8	279.6	-51.6	565	2.3				
PE20	granodiorite	71°42'40";16°32'15"	0.9	0.072						8	136.7	46.1	95	5.7				
PE26	gabbro	71°42'48";16°28'28"	3.2	0.098						7	287.2	-47.9	14	16.7				
								Ilo										
PE28	andesitic flow	71°23'00";17°25'25"	0.03	0.037	3	335.1	-39.9	160	9.8	7	187.3	-19.5	95	6.2	297	63	182.6	39.7
PE29	andesitic flow	71°23'03";17°25'29"	0.06	0.038	5	336.2	-34.6	91	8.0	7	178.5	-31.7	80	6.8	297	70	178.6	31.5
PE30	andesitic flow	71°23'06";17°25'37"	0.07	0.015	10	332.0	-39.1	280	2.9	10	185.2	-14.5	302	2.8	290	70	176.0	52.6
PE31	rhyolitic flow	71°22'55";17°25'57"		0.004	5	318.8	-45.3	29	14.6	7	187.8	-7.6	48	8.8	287	61	182.0	52.3
PE32	andesitic dike	71°22'55";17°25'57"	2.3	0.073						5	181.6	35.1	238	5.0				
PE34	diorite	71°21'40";17°31'38"	3.2	0.420						9	353.4	-25.3	595	2.1				
								La Yarada										
PE36	andesitic flow	70°43'35";18°06'51"	0.1	0.007	7	325.5	-43.6	118	5.6						212	30		
PE37	ignimbritic flow	70°43'35";18°06'51"	0.01	0.002						7	155.8	52.4	57	8.0	212	30	204.8	70.0
PE38	andesitic flow	70°43'20";18°07'18"	0.5	0.059	10	330.4	-31.3	66	6.0						267	20		
PE40	andesitic sill	70°44'08";18°05'58"	0.07	0.003	6	338.8	-40.2	211	4.6						245	15		
PE42	andesitic dike	70°45'10";18°05'21"	1.5	0.072	8	333.9	-27.9	107	5.4						132	22		
PE43	dacitic flow	70°45'10";18°05'21"	0.9	0.067						8	276.8	27.5	319	3.1	132	22	270.2	13.7
PE44	dacitic flow	70°44'50";18°05'23"	1.9	0.190	5	328.8	-31.1	161	6.0	9	275.1	31.7	165	4.0	127	40	263.6	6.5
PE45	andesitic sill	70°45'30";18°05'25"	0.05	0.002	9	332.6	-41.7	198	3.7	6	158.0	52.7	150	5.5	247	20	159.0	72.7
PE48	andesitic dike	70°46'35";18°03'58"	1.5	0.130	9	334.9	-39.3	121	4.7						350	20		
PE49	andesitic dike	70°46'35";18°03'58"	0.06	0.005	8	317.5	-48.8	108	5.4						350	20		
								Punta de Bombon										
PE50	tonalite	71°40'40";17°01'20"	3.8	0.081						8	158.1	40.1	150	4.5				
								Lake Titicaca - Bolivia										
BO-27	Permian sed	68°51'00";16°12'00	0.04	0.01						7	72.6	21.7	31	10.9	115	40	95.6	43.3
BO-29	Triassic sed	68°52'00";16°12'00	0.03	0.006						9	268.5	-3.0	137	4.4	110	63	282.7	-20.5
BO-30	Triassic sed	68°52'30";16°12'00	0.03	0.006						9	273.3	-16.9	84	5.7	115	35	287.1	-26.2

χ, geometric mean susceptibility (0.01SI); J_{NRM}, geometric mean intensity of magnetization in Amperes per meter; *N*, number of samples used in the calculation of the mean direction; *D*, *I*, mean declination and inclination; *k*, Fisher's precision parameter; α_{95}, semi-angle of confidence. Secondary IS is component B in-situ; Primary TC is component A tilt corrected.

al., 1974; *Moore*, 1984; *Beckinsale et al.*, 1985; *Mukasa*, 1986].

PETROGRAPHY

In Jurassic rocks, the lithology of the rocks sampled for paleomagnetism corresponds to basaltic to dacitic flows, andesitic dikes (or sills), and one site of red bed sediments. Twenty-six samples were selected for petrographic studies.

In the Chala area, at all sites except sites PE09 and PE10 the clinopyroxene and plagioclase from the basaltic flows are generally preserved. Hematite (site PE01) is observed as a product of deuteric oxidation (classes 4 and 5 as defined by *Haggerty* [1976]). There is no convincing evidence for opaque granulation or opaque pseudomorphs by black titanite, characteristic of burial metamorphism [*Ade-Hall et al.*, 1971]. However, at sites PE09 and PE10, the clinopyroxene is replaced by chlorite, calcite, hematite, rutile, and titanite whereas the plagioclase is altered to albite, white mica, and, locally, pumpellyite and epidote. Therefore these rocks have probably suffered a nondeformational, low-grade alteration of hydrothermal-burial type (spilitization) in the zeolite and prehnite-pumpellyite facies as is well documented in adjacent regions [*Palmer et al.*, 1980b; *Aguirre and Ofler*, 1985; *Aguirre*, 1988].

At Ilo and La Yarada, the magmatic paragenesis is generally not preserved. Plagioclases, clinopyroxenes, and olivines are completely transformed to actinolite, chlorite, calcite, epidote, and hematite associations. Textures of primary or high-temperature deuteric alteration were obliterated in opaque minerals. Hematite occurs as homogeneous large crystals and as minute disseminated phases in pseudomorphosed phenocrysts and in the groundmass. Near Ilo, numerous hematite-bearing veins crosscut the Chocolate Formation. Tourmaline occurs as a hydrothermal product in the volcanic breccias. These features indicate that contact metamorphism has occurred in the Ilo and La Yarada areas. This contact metamorphism is produced by the Jurassic and Cretaceous plutonism and is superimposed onto the low-grade alteration of hydrothermal-burial type.

In intrusive rocks, optical observations indicate that primary magnetite is the main magnetic phase except at La Planchada near Ocoña (sites PE15 and PE16), where hematite with ferrian ilmenite lenses results from the transformation of primary ilmenite [*Haggerty*, 1976]

PALEOMAGNETIC RESULTS

Paleomagnetic Sampling

The paleomagnetic sampling was made at different localities along the southern coast of Peru with 47 paleomagnetic sites and about 430 cores. Three additional sites from Late Permian to Early Triassic red bed sediments were also sampled near the Straits of Tiquina on the Bolivian side of Lake Titicaca. Paleomagnetic results have been obtained from 39 sites out of the 50 sampled sites. The paleomagnetic results as well as the locations of the corresponding sites are given in Table 2.

Generally 8 to 12 cores were drilled at each site using a portable gasoline powered core drill. Each site consisted of one homogeneous geological unit (either a single flow, a dike, a sill or an outcrop of a few tens of meter wide, of an intrusive body or sediments). This procedure enables us to check the consistency of the paleomagnetic results for a single site, but the primary magnetizations of sites corresponding to single flows or dikes do not average out geomagnetic secular variation.

Near Chala, 11 sites were sampled in the basaltic to dacitic Jurassic units, and one site (PE12) of red sandstone interbedded in these volcanics was sampled. Site PE14, a few kilometers north of Chala, corresponds to a plutonic intrusion attributed to the superunit Linga by *Olchauski* [1980]. Two sites were collected in the La Planchada monzodiorite intrusion near Ocoña. Seven sites were drilled in different units of the Peruvian coastal batholith from the Arequipa area. Five sites of the Chocolate volcanic formation were sampled near Ilo. Three sites in nearby intrusive formations were also sampled, but only one site (PE34) in the Punta Coles diorite has a stable remanent magnetization. Fourteen sites were collected in andesitic to rhyolitic flows, sills, and dikes in the area of La Yarada, west of the city of Tacna.

Measurements of the remanence have been performed on a DSM1 Schonstedt spinner magnetometer. Alternating field (AF) demagnetizations were performed with a GSD1 Schonstedt demagnetizer. The residual field in the furnace used for the thermal demagnetization was less than 10 nT.

Magnetic Properties

Large variations in magnetic properties are observed between sites as indicated by the geometric mean values per site of the magnetic susceptibility and the intensity of the natural remanent magnetization (NRM) listed in Table 2. In particular, low intensities of magnetization (about 0.01 A m^{-1}) and low susceptibility values are observed in several sites from the volcanic units.

Isothermal remanent magnetization (IRM) acquisitions have been performed on several samples in order to assess the relative importance of hematite in the samples (Figure 3). Saturation values for samples carrying only magnetite are generally obtained for field values between 0.1 and 0.3 tesla. (for example; sample PE05-07). The maximum applied laboratory field was only 0.8 T, and because saturation fields for hematite can be as high as 6 T, the steady increase in saturation magnetization versus laboratory fields up to 0.8 T is interpreted as evidence for hematite. Magnetite and hematite are both present in samples of volcanics from Chala and Ilo (Figure 3). On the other hand, IRM experiments indicate that magnetite is the main magnetic carrier for sills and dikes from La Yarada (Figure 3). It is important to remember that saturation magnetization for magnetite is over 90 times that of hematite, and thus magnetite is more easily recognized during IRM acquisition.

Magnetic properties of intrusive plutonic rocks were due mostly to magnetite; only the La Planchada intrusion near Ocoña was atypical with hematite being the only magnetic carrier (sample PE16-05, Figure 3).

Strong-field thermomagnetic experiments performed on a few samples were successful only for those with high enough susceptibility and indicated that magnetite was the only magnetic carrier for these samples (Figure 4). There is no evidence for Ti-rich titanomagnetite or titanomaghemite in these rocks. Since we did not try to isolate the magnetic minerals before the experiment, it was not possible to detect hematite with the vertical Curie balance.

Because of the occurrence of hematite, progressive thermal demagnetizations were generally very efficient in isolating different components of magnetization.

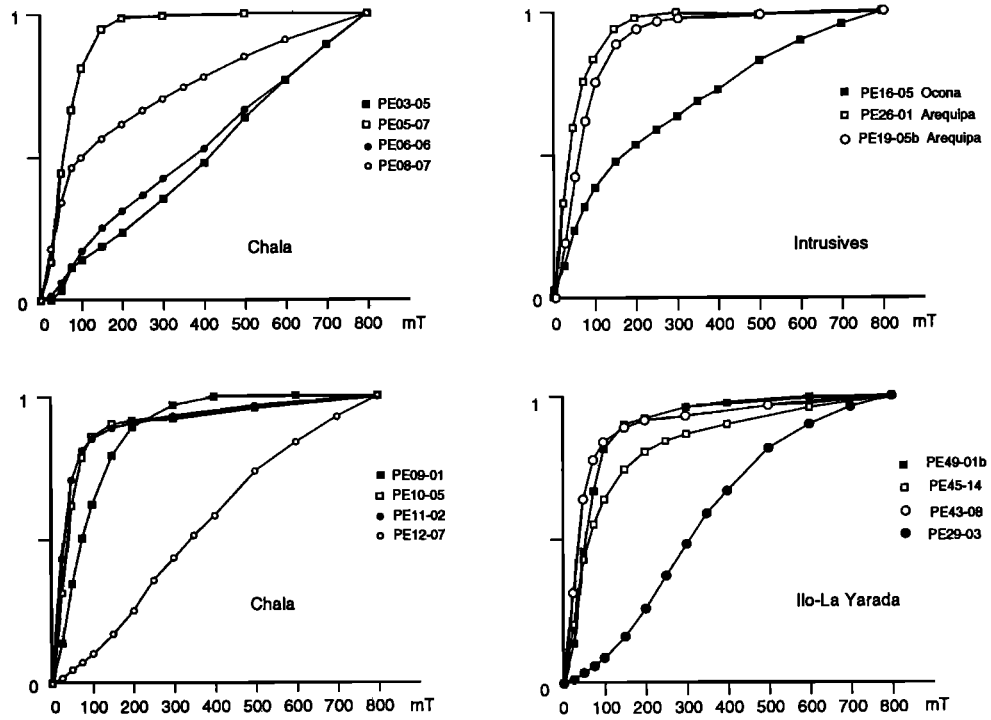


Fig. 3. Isothermal remanent magnetization (IRM) acquisition curves for 15 samples from different sites. Each curve was normalized by the IRM value at 0.8 tesla.

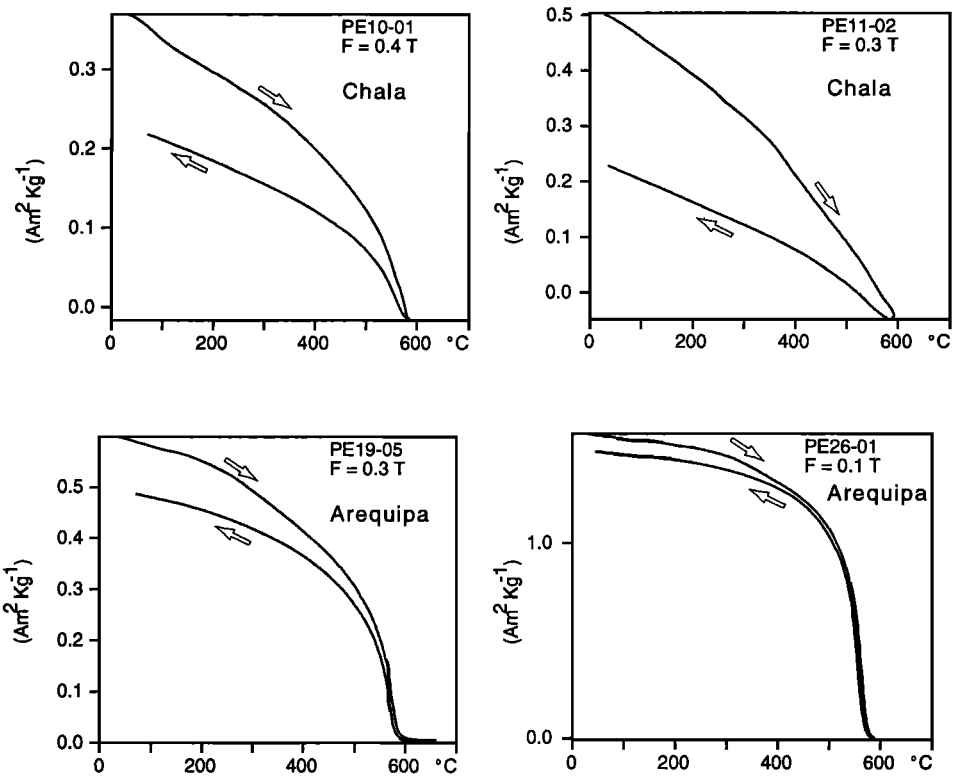


Fig. 4. Examples of thermomagnetic experiments with applied fields from 0.1 to 0.4 tesla.

Directions

The paleomagnetic directions were determined by least squares fit analysis. The demagnetization interval was chosen for each sample with the help of orthogonal demagnetization diagrams. The least squares fit was performed on a segment with at least four demagnetization steps leading to a significant magnetization vector generally greater than 25% of the initial natural remanent magnetization (NRM). The standard deviation associated to the least squares fit was usually lower than 10°. Directions at more than twice the angular standard deviation

were rejected from the site mean calculation. Although 8 to 12 cores were collected at each site, characteristic magnetizations (either secondary or primary) were not isolated in all samples. Typical examples of demagnetization behavior are shown for each rock type. The magnetization vector removed at 150°C or 10 mT is considered as a viscous or spurious magnetization and was discarded.

Chala. Both normal and reversed polarities are often found within individual samples (sample 88PE1005A, Figure 5). Several sites from the volcanic units have the same component of magnetization of normal polarity with a declination deflected to the northwest by 30° to 40° with respect to the present geographic north. This component of magnetization is usually defined in the temperature range 150°-450°C (and referred to in this text as component B) as indicated by examples from Figure 5 (sample PE0305B). This direction of magnetization is also the only one isolated in sample PE0903A (Figure 5) with unblocking temperatures widely distributed between 150° and 560°C. Component B was also the only one removed during demagnetization of samples from the site of red bed sediments (PE12); the demagnetization paths did not converge through the origin, and the magnetization at 660°C was highly variable between the different cores of the site, but a single component of magnetization was clearly identified on the orthogonal plots (sample PE1201A, Figure 5) and the site mean direction is very well defined ($\alpha 95<3°$). Because the tectonic corrections are the same for most sites, it is difficult to perform a fold test in order to clearly identify a primary origin for a component of magnetization. Component B has been identified in nine sites (Table 3 and Figure 6). By applying the tilt correction, the Fisher's precision parameter decreases by a factor of 2 from a value of k=139 to k=70 which suggests a secondary origin for this component (Figure 6 and Table 3). Moreover, the relative high value of the precision parameter before the tilt correction supports the hypothesis of a remagnetization because this process can average geomagnetic secular variation. While there is evidence for a low-grade metamorphism at sites PE09 and PE10 where magnetite is the magnetic carrier, the remagnetizing process is poorly understood at the other sites (volcanics and the red beds) where the secondary magnetization appears to be carried by hematite.

A characteristic component (component A) of magnetization with high unblocking temperatures is observed in seven sites from the volcanic flows. Examples of this type of behavior are given by samples PE1102A and PE0402A (Figure 5). These samples have a component of magnetization carried by magnetite with high unblocking temperatures above 500°C

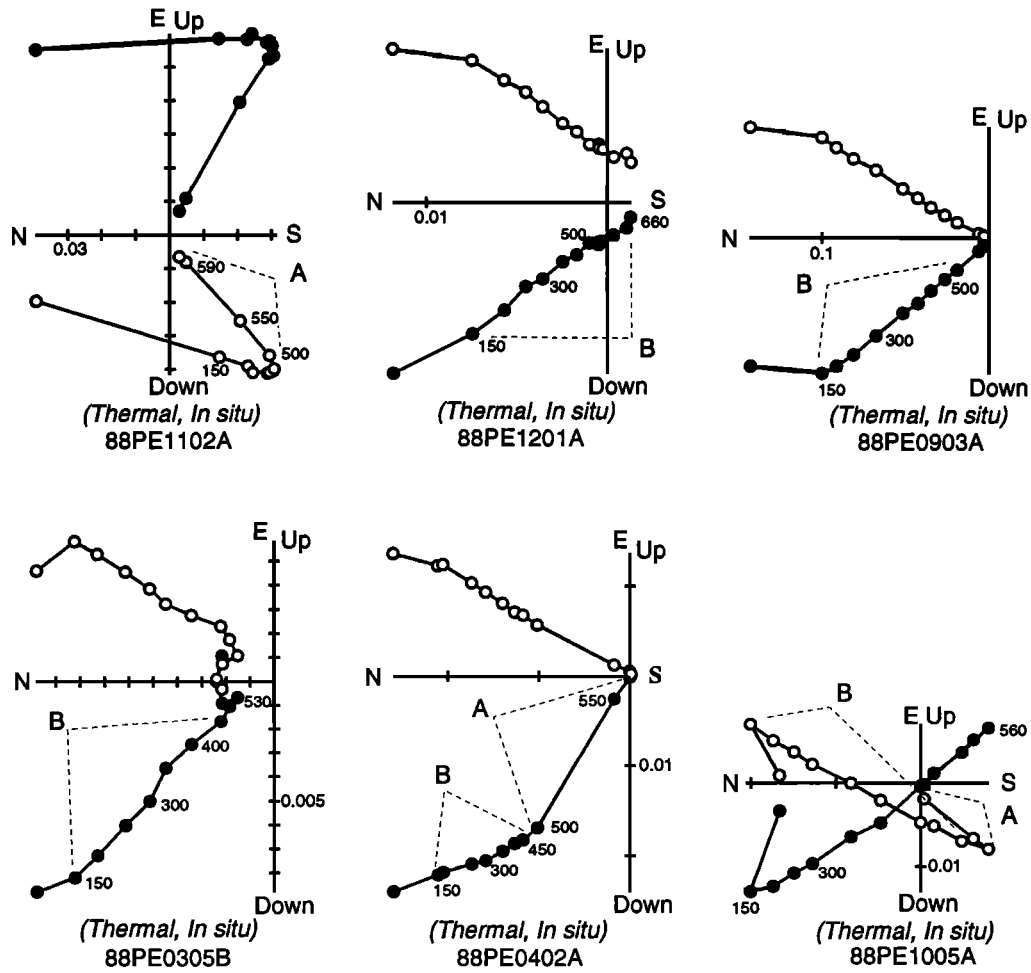


Fig. 5. Typical orthogonal diagrams of thermal demagnetization from six sites of the Chala area. Open circles correspond to the projection into the vertical plane while solid circles are the projection of the horizontal components. Temperatures are given in degrees Celcius, and the intensity of magnetization is in amperes per meter. Dashed lines illustrate the characteristic components of magnetization A and B.

TABLE 3. Area Mean Paleomagnetic Results

Locality	In Situ N	D	I	k	α_{95}	After tilt correction N	D	I	k	α_{95}
Chala (secondary)	9	**324.9**	**-25.1**	**139**	**4.4**	9	309.7	-47.2	70	6.2
Chala (primary)	6	305.0	-21.7	16	17.1	6	289.8	-35.7	14	18.7
Chala (primary)*	8	304.0	-19.3	20	12.7	**8**	**290.5**	**-31.3**	**13**	**16.0**
Ocoña (plutonic)	**2**	**270.8**	**-48.3**	-	-					
Arequipa (intrusions)	**4**	**299.5**	**-47.0**	**37**	**15.4**					
Ilo (secondary)	**4**	**330.9**	**-39.9**	**119**	**8.5**	*4*	259.7	-44.1	157	7.4
Ilo (primary)	4	4.9	18.3	55	12.4	4	**359.9**	**-44.1**	**60**	**12.0**
La Yarada (secondary)	**8**	**330.6**	**-38.1**	**87**	**6.0**	*8*	330.9	-42.3	12	16.7
Tiquina-Bolivia (primary)	3	265.0	-14.0	32	21.9	**3**	**282.2**	**-30.1**	**40**	**19.7**

N is number of sites used in the calculation of the mean direction. *D* and *I* are mean declination and inclination; *k* is Fisher's precision parameter; α_{95} is semiangle of confidence. The characteristic results before or after tilt correction are identified in bold.
* dike PE05 and site PE14 included

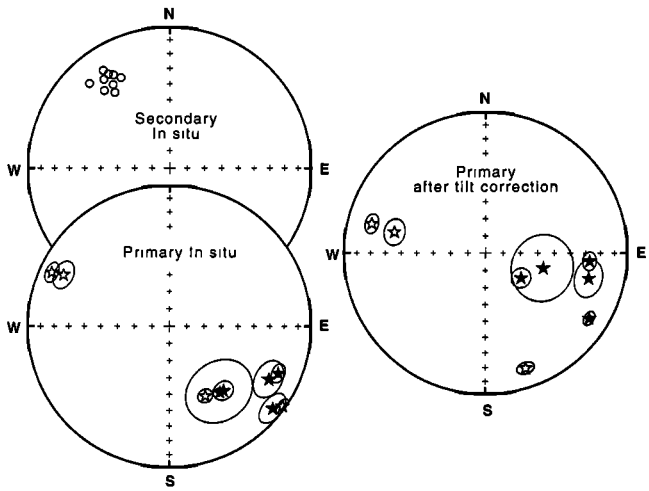


Fig. 6. Equal-area projections of site mean paleomagnetic results for the Chala area. The upper left diagram illustrates the distribution of the secondary magnetizations (component B) shown by circles; the cones of confidence at 95% were omitted for clarity. The paleomagnetic directions interpreted as primary magnetizations are shown with stars and associated ovals of confidence in in situ coordinates and after tilt correction (open (solid) symbols are projection in the upper (lower) hemisphere).

which may represent the primary magnetization of the rocks. The directions observed for the volcanic flows (Table 3) are in agreement with those observed for the dike (PE05) and for one site (PE14) of the intrusive stock (the magnetization of the other site PE13 was unstable). The paleomagnetic direction recorded at site PE07 is at more than twice the standard deviation around the mean; it can be interpreted as an intermediate direction and is not included in the mean calculation. Although the tilt correction does not reduce the scatter in these data, the high-temperature stability of the magnetization and the antipodal distribution of normal and reversed site mean directions (Figure 6) support the hypothesis of a primary magnetization. The mean direction from eight sites has a declination of about 290° and an inclination of -30° after applying a tilt correction for the volcanics based on the interbedded sediments but not for the granite stock for which no tilt correction is available.

Ocoña. Two sites were drilled in a plutonic intrusion of La Planchada near Ocoña. The magnetic carrier is mostly hematite as indicated by the demagnetization diagram (sample PE1602B, Figure 7). Both sites have a stable magnetization with an east declination and a positive inclination which is interpreted as a reversed polarity.

Arequipa. Seven sites were drilled in different units of the Peruvian coastal batholith around Arequipa. However, three sites did not provide consistent results because the magnetization was unstable upon stepwise demagnetization. Examples of successful demagnetizations are given in Figure 7. Both polarities are recorded and are almost antipodal (Figure 8); this constitutes a reversal test, and the mean direction is certainly representative of a geocentric axial dipole field. No tilt correction is available for these intrusive rocks. However, the mean inclination is not very different from the expected inclination, and normal and reverse site mean directions are observed at different sites several kilometers apart. Thus the large deviation of the declination is best interpreted by counterclockwise rotations about a vertical axis rather than deflections of the declinations due to block tilting. This interpretation is further supported by results from the study of the anisotropy of magnetic susceptibility (see next chapter).

Ilo. A complex multicomponent magnetization was found during thermal demagnetization of samples from the volcanic flows. A secondary component equivalent to component B determined at Chala is identified in the temperature range 150-450°C (Figures 9 and 10). Another component carried by pure magnetite is found in the range 500°-560°C (Figures 9 and 10) and appears to be characteristic of the sites. In site PE30, the magnetization carried by hematite with unblocking temperature higher than 580°C (sample 88PE3005, Figure 9) was not homogeneous within different cores of the same site. The tilt correction applied to the direction isolated in the temperature range 500°-560°C yields a reversed magnetization and a positive inclination in better agreement with the expected inclination for these formations (Figure 10). This strongly suggests that this component is a primary magnetization. A reversed magnetization was also found in a dike (PE32) which may postdate the folding. Of the three sites sampled in plutonic rocks, only one (see sample PE3406A, Figure 9) provided a clear paleomagnetic signal with a normal polarity.

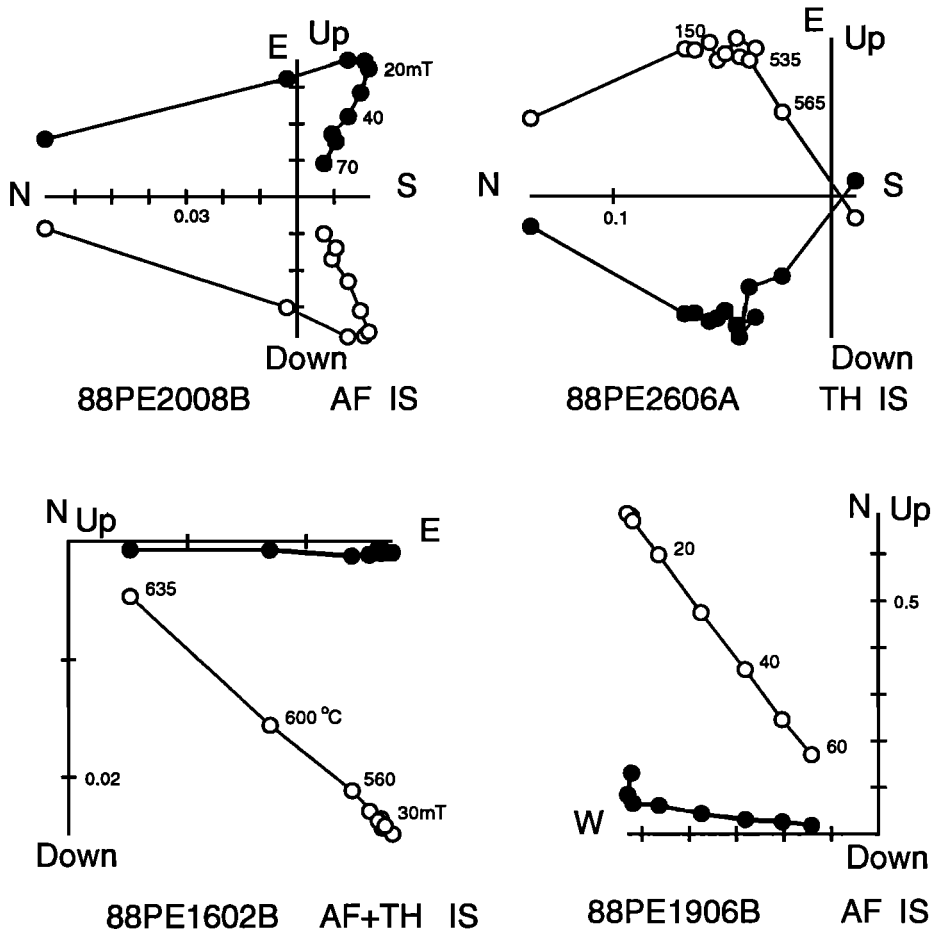


Fig. 7. Orthogonal diagrams of demagnetization for samples from plutonic rocks from Ocoña and Arequipa (Same conventions as in Figure 5). Demagnetization was first attempted by alternating field (AF) on sample PE1602B and completed thermally.

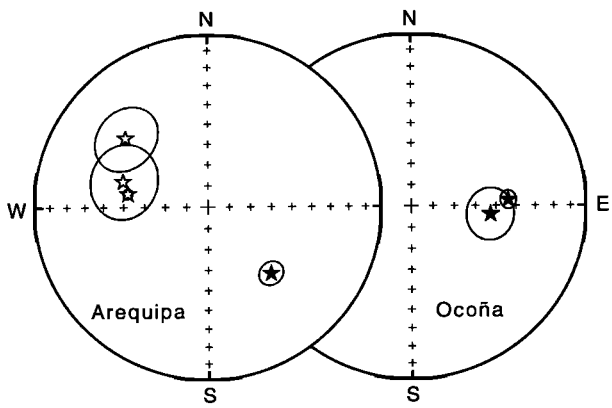


Fig. 8. Equal-area projection of site mean paleomagnetic results from plutonic rocks from Arequipa and Ocoña (same conventions as in Figure 6).

While the declination of the secondary magnetization is deflected to the NW, results after tilt correction for the primary magnetization correspond to a N-S orientation, in good agreement with uncorrected results from the dike and the intrusive. The difference between the secondary component (which has recorded a counterclockwise rotation) and the primary magnetization almost N-S suggests that a local clockwise rotation occurred before the time of remagnetization. These preliminary data may indicate a complex tectonic history for this area, and a more detailed sampling is certainly needed to assess the chronology of the different events.

La Yarada. A characteristic magnetization similar to component B defined at Chala and Ilo was found at eight sites. Some examples of demagnetization diagrams which show this component of magnetization are given in Figure 11. An interesting feature was found at site PE45. Site PE45 consisted of a 20 m thick sill interbedded in sediments. We sampled the interior, the bottom part of the sill and the sediments which were baked by the intrusive sill. A reversed component with high unblocking temperature (sample 88PE4505A, Figure 11) was found in the baked sediments and the oxydized bottom part of the sill, while the interior part of the sill yielded only the B component of normal polarity with low to intermediate unblocking temperatures. The site mean results plotted in Figure 12*a* indicate good internal consistency for the B component before tilt correction. The various attitudes of the bedding planes measured in interbedded sediments enable a fold test to be made. The large decrease in the Fisher's precision parameter k from 87 to 12 (Table 3) demonstrates that component B is a postfolding magnetization.

The other characteristic magnetizations are more difficult to interpret. A characteristic direction (west declination and positive in situ inclination, (Figure 12*b*) carried by magnetite with unblocking temperature greater than 500°C was found at sites PE43 and PE44. Although the sites are a few hundred meters away from one another, the direction of the strike of the bedding plane and the locations of the sites indicate that these

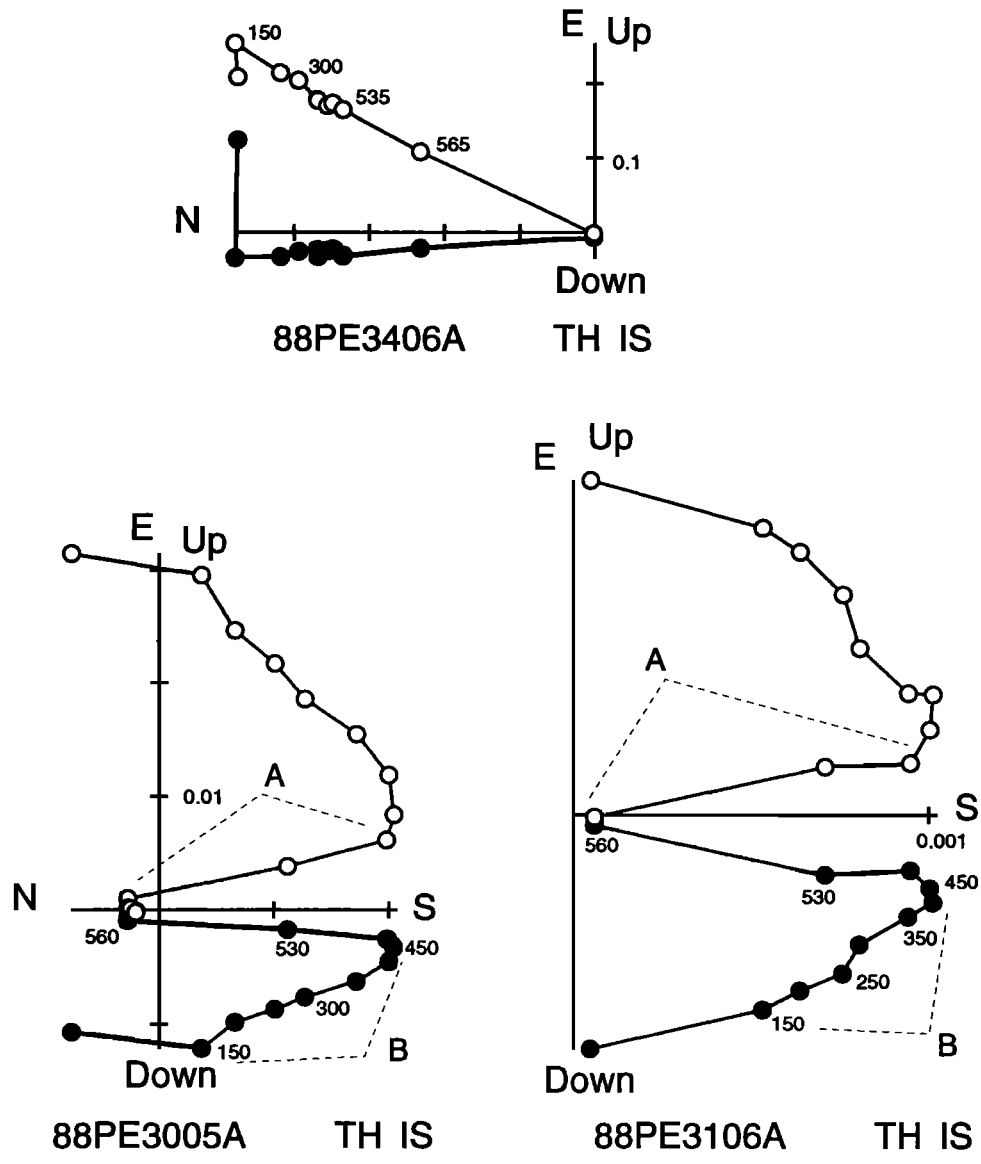


Fig. 9. Demagnetization diagrams from three sites from the Ilo area. A secondary component (B) with a N-NW declination is clearly identified in the temperature range 150°-450°C for samples PE3005A and PE3106A. Another component (A) with south declination and shallow negative inclination (in situ) is isolated in the interval 500°-560°C. Sample 88PE3406A illustrates the demagnetization behavior of samples from the intrusive stock (site PE34) (same conventions as in Figure 5).

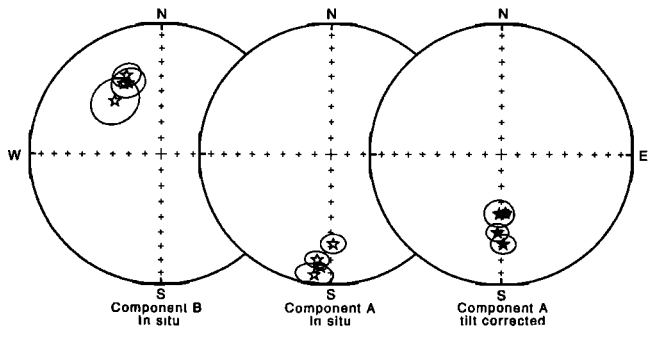


Fig. 10. Equal-area projection of in situ site mean directions for Jurassic volcanics from Ilo. After tilt correction (right), the component A has a positive inclination (same conventions as in Figure 6).

two sites are stratigraphically close. Because this atypical direction is found within only two sites, it can be interpreted as an intermediate paleomagnetic direction recorded by successive flows of the same age; thus we will not try to interpret this direction in a tectonic context. Another characteristic direction of reversed polarity is also found at site PE37 and in the oxydized materials at site PE45 (Figure 12*b*). The question whether to apply the tilt correction to the assumed primary magnetization recorded in the baked sediments remains unanswered since the intrusion of the sill may also postdate the folding.

Punta de Bombon. Only one of the two sites sampled in two different units of the batholith provided accurate paleomagnetic results. The reversed polarity paleomagnetic direction is consistent with those observed at La Yarada.

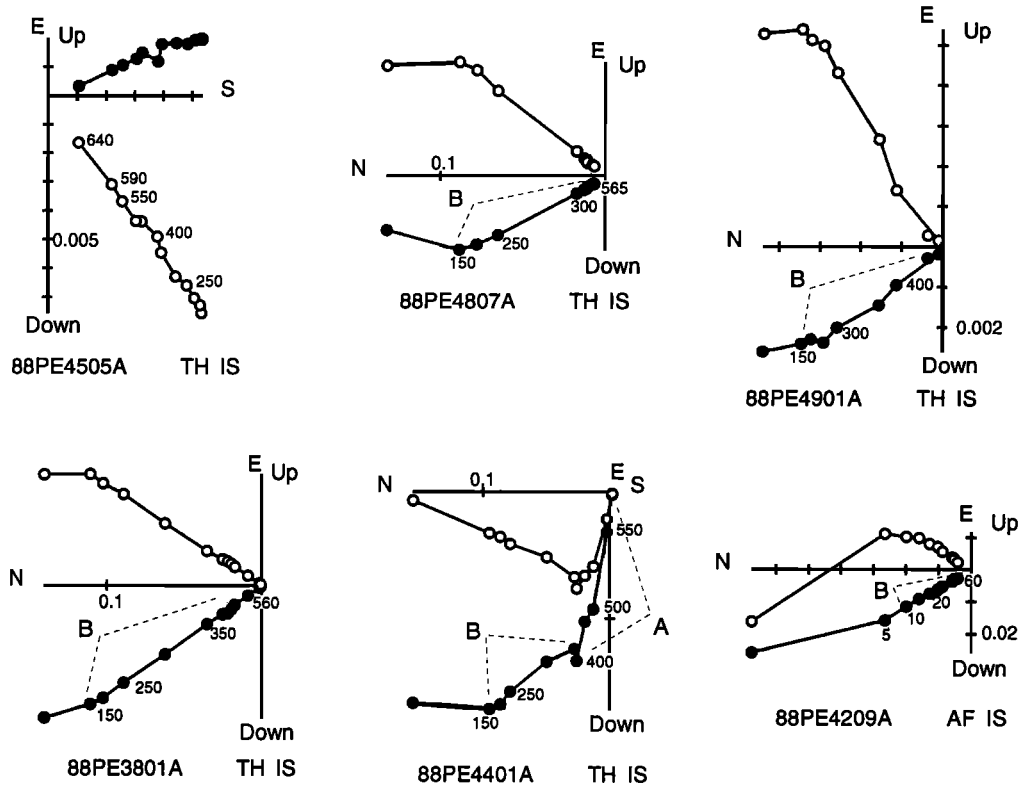


Fig. 11. Orthogonal demagnetization diagrams for samples from LaYarada: (same conventions as in Figure 5).

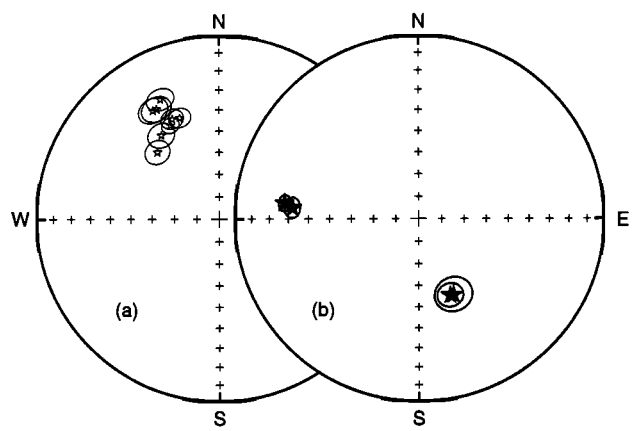


Fig. 12. Equal-area projection of site mean paleomagnetic results from LaYarada; (*a*) component B site-mean results and (*b*) component A site-mean results (same conventions as in Figure 6).

Permian sediments: Lake Titicaca. Thermal demagnetizations of samples from fine-grained siltstone revealed a well-defined characteristic magnetization (Figures 13*a* and *b*). The sign of the inclination after tilt correction indicates that the stratigraphically lowest site (BO27) has a reversed polarity while the two other sites (BO29-BO30) have a normal polarity. The thermal unblocking temperatures indicate that hematite is the main magnetic carrier. The change in polarity from reverse to normal suggests that the sediments were deposited after the Kiaman long-reverse chron. The tilt correction is not significant enough to ensure that the magnetization is of primary origin. However, the mean inclination is steeper after tilt correction and in better agreement with the expected inclination (Tables 3 and 6). Moreover, the observation of normal and reversed polarities support the hypothesis of a primary origin.

Summary of Paleomagnetic Results

All mean results are summarized in Table 3. A characteristic magnetization (component B) is identified in the Jurassic volcanics at three different localities (Chala, Ilo, and La Yarada) with a well-defined mean virtual geomagnetic pole (59.5°N, 190.1°E, N=21, k=111, $\alpha 95$=3.0). Evidence for low-grade metamorphism, low to intermediate unblocking temperatures, and negative fold tests indicates that this magnetization is secondary. This magnetization is different from a recent viscous overprint acquired during the normal Bruhnes chron. The remagnetization process should be related to one of the major tectonomagmatic events occurring in southern Peru. The overthrust of the Arequipa massif in early Late Cretaceous [*Vicente* et al., 1979; *Vicente*, 1989] and the intrusion of the batholith constitute the best sources for a remagnetizing event. We believe that the remagnetization may be contemporaneous or younger than this early phase of Andean deformation. Thus we will assume a Late cretaceous age for this magnetization.

Antipodal normal and reversed site mean directions and high unblocking temperatures support the hypothesis that the magnetizations isolated in intrusives stocks and component A in Jurassic volcanics are of primary origin. Mean directions from primary magnetizations from the Chala, Ocona, and Arequipa areas correspond to counterclockwise rotated declinations (Tables 2 and 3). We interpret these results from

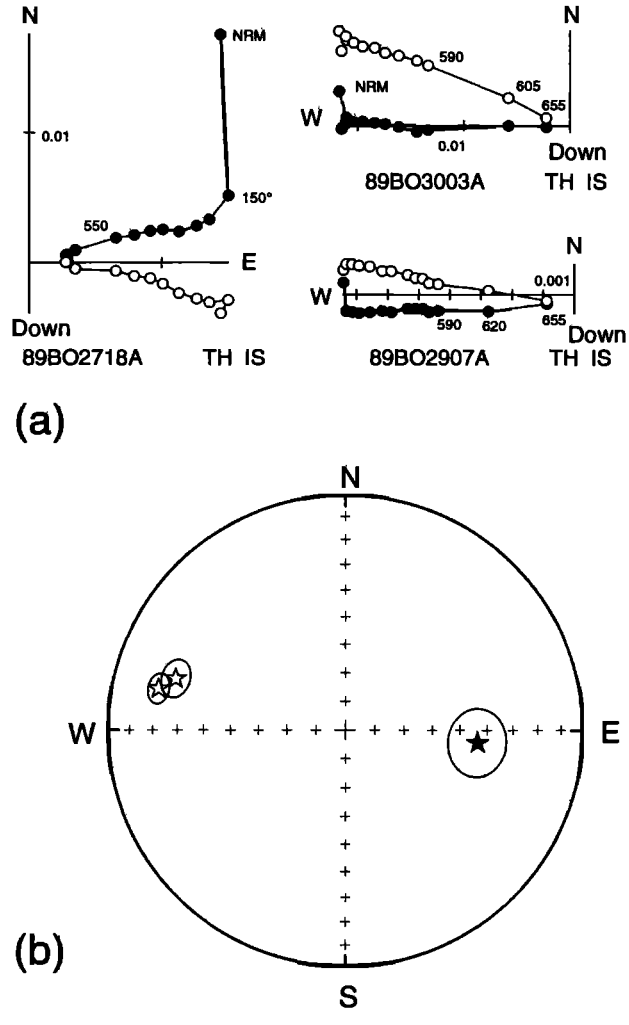


Fig. 13. (*a*) Orthogonal demagnetization diagrams for samples from Late Permian to Early Triassic sediments near the Lake Titicaca. (*b*) Equal-area projection of site mean paleomagnetic results.

primary magnetizations as local block rotations prior to the remagnetization event. The large counterclockwise rotations seen in the northern areas of Chala, Ocona, and Arequipa are not observed at Ilo and La Yarada. However, the limited number of sites at Ilo and La Yarada for which a characteristic primary direction is isolated precludes a clear interpretation of these data. Preliminary results from Bolivia also suggest large counterclockwise rotation.

Strain-Induced Anisotropy of Magnetic Susceptibility in Intrusive Rocks

The low-induction magnetic susceptibility anisotropy of a rock is described as a symmetric tensor of second rank; it can be represented graphically as an ellipsoid with principal axes k_{max}, k_{int} and k_{min}. When k_{int} is closer to k_{max} than k_{min}, the ellipsoid takes an oblate shape and the magnetic fabric defines a foliation. In the other case when k_{int} is closer to k_{min}, the ellipsoid is prolate with a magnetic lineation.

The anisotropy of magnetic susceptibility (AMS) was measured at each site from plutonic stocks in order to assess (1) the importance of the deformation, (2) its effect upon the remanence direction and (3) its tectonic significance. Measurements were made with the Digico apparatus at the University of Rennes, and all results are summarized in Table 4. Most susceptibility tensors had an oblate shape with a steep foliation plane. The degree of anisotropy was up to 15%.

Laboratory experiments [*Cogné*, 1987] demonstrated that the thermoremanent magnetic (TRM) anisotropy deviation tensor can be deduced from the AMS tensor whose eigenvalues ratios are approximately equal to the square roots of those of the TRM tensor. Assuming that the magnetic fabric was acquired prior to the acquisition of the magnetization, the measured paleomagnetic directions can be corrected by inverting the calculated TRM tensor. The estimated induced deviations (Table 4) of the site mean paleomagnetic directions are small (lower than 5°, except for site PE17) with respect to the amount of rotation recorded by these rocks and can be neglected in further analyses.

TABLE 4. Anisotropy Parameters for Intrusive Rocks from Southern Peru

Site	*N*	$k$max	*D*	*I*	k_{int}	*D*	*I*	k_{min}	*D*	*I*	k_{max}/k_{int}	k_{int}/k_{min}	%An	Dec	Inc	Dec'	Inc'
PE13	8	1.059	318.3	34.6	1.025	1.8	-46.4	0.916	65.3	23.0	1.033	1.119	15.6				
PE14	7	1.047	142.0	20.3	1.033	256.9	48.8	0.920	37.5	34.1	1.014	1.122	13.8	128.6	7.2	127.6	7.3
PE15	10	1.031	92.2	6.0	0.989	357.0	40.6	0.980	9.1	-48.7	1.042	1.009	5.2	96.1	52.5	97.2	55.1
PE16	5	1.024	115.4	17.4	1.001	18.0	22.4	0.974	240.0	61.0	1.022	1.028	5.1	86.4	43.9	85.3	45.5
PE17	8	1.059	72.6	70.5	1.013	104.0	-16.8	0.927	11.1	-9.6	1.045	1.092	14.2	310.4	-38.2	317.2	-34.9
PE18	10	1.049	303.5	71.9	1.023	115.8	17.9	0.928	206.5	2.3	1.025	1.103	13.1				
PE19	8	1.045	35.6	59.5	1.001	125.2	-0.2	0.953	215.1	30.5	1.044	1.050	9.7	279.6	-51.6	283.8	-50.7
PE20	17	1.029	170.0	41.4	1.000	1.6	48.0	0.971	85.1	-5.8	1.028	1.030	6.0	136.7	46.1	133.5	45.4
PE25	5	1.029	152.8	69.9	1.021	304.9	18.0	0.950	37.8	8.8	1.007	1.075	8.3				
PE26	14	1.018	334.7	70.1	0.997	316.0	-19.0	0.985	48.1	-5.9	1.021	1.013	3.4	287.2	-47.9	287.2	-46.7
PE27	9	1.029	11.9	69.3	1.007	323.1	-14.0	0.964	236.9	15.0	1.022	1.044	6.7				
PE33	6	1.027	359.2	-30.1	1.014	45.1	50.2	0.959	283.7	23.5	1.012	1.058	7.1				
PE34	7	1.061	141.9	57.1	1.000	50.1	1.1	0.938	319.4	32.8	1.061	1.067	13.2	353.4	-25.3	353.3	-19.8
PE35	9	1.006	204.2	0.0	1.005	114.2	-27.3	0.988	114.2	62.7	1.001	1.018	1.9				
PE50	9	1.011	134.2	-34.2	1.000	277.8	-49.9	0.988	31.1	-18.4	1.011	1.013	2.4	158.1	40.1	159.2	40.3
PE51	7	1.015	14.5	-26.4	1.007	283.6	-1.7	0.977	10.3	63.5	1.007	1.031	3.8				

N is number of samples used to determine the mean normalized susceptibility tensor; k_{max}, k_{int} and k_{min} are the eigenvalues with their respective eigenvectors (*D*,*I*) in in situ coordinates; %An is $((k_{max}/k_{min})-1)\times100$; Dec and Inc are mean paleomagnetic directions; Dec' and Inc' are corrected paleomagnetic directions.

AMS results from the Arequipa area (Figure 14) indicate a foliation plane mostly NW-SE (except site PE20) and vertical. This direction is subparallel to the main direction of deformation (the Cincha-Lluta overthrust). The subvertical magnetic foliation and the slight vertical magnetic lineation recorded at these sites are probably related to emplacement tectonics of diapiric intrusions. These AMS results also suggest that large block tilting is unlikely.

The counterclockwise rotation recorded by the remanence for the Arequipa area indicates that the structural directions have also been rotated. Using the paleomagnetic directions, we can apply a rotation to the foliation plane. This gives a state of strain similar to the one observed at Ilo. A more extensive study of the AMS of the batholith (which is the most easy way of investigating the state of strain) would result in a better understanding of the relationship between plutonic intrusions in the Andes and the tectonics of the region.

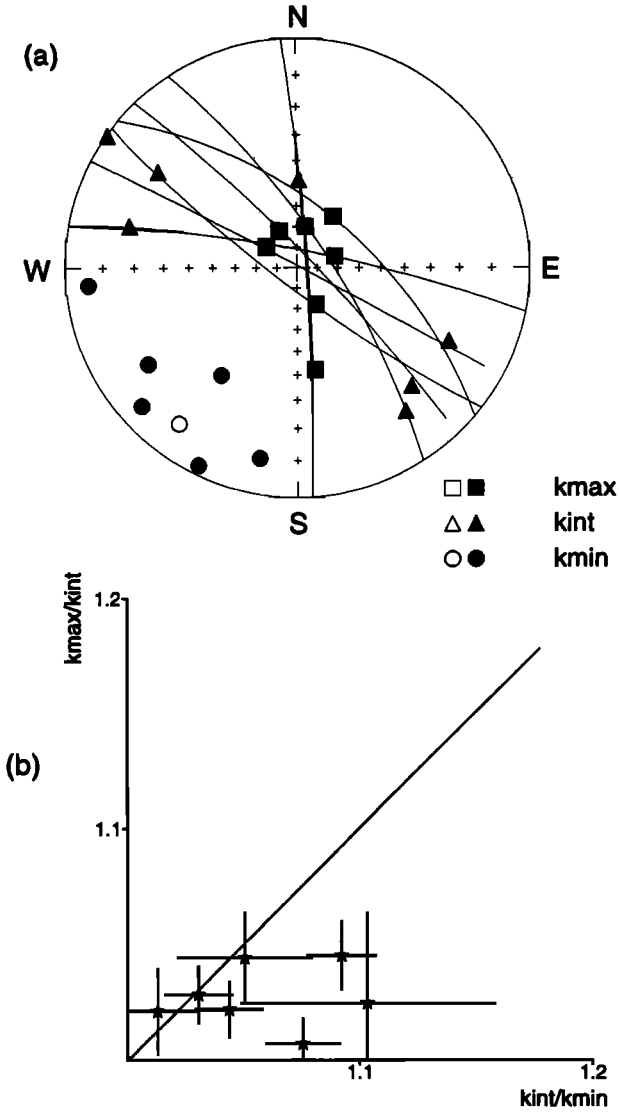


Fig. 14. (*a*) Stereographic projections of principal axes of anisotropy of magnetic susceptibility (AMS) tensors with their associated foliation planes for six sites in intrusions from Arequipa. (b) Anisotropy plot of site mean AMS data.

A Reference Apparent Polar Wander Path Curve for South America.

The distribution of late Mesozoic and Cenozoic paleomagnetic poles from South America exhibits an elongated pattern which was interpreted by *Valencio et al.* [1983] as a result of the drift of the South American plate. More recently, *Beck* [1988] has suggested that most of the dispersion in the South American paleomagnetic poles is due to differential rotations in the Andes while paleomagnetic poles for the stable parts of South America are not much different from the present day rotation axis of the earth. Thus *Beck* [1988] has proposed a single mean paleomagnetic pole of reference for the whole Cretaceous time period instead of an apparent polar wander path (APWP) curve based on few sites.

Besse and Courtillot [1991] have provided a synthetic APWP for the last 200 m.y.. The APWP from Africa was rotated to South America (SA) by progressive closure of the South Atlantic ocean using rotation parameters from *Cande et al.* [1988] and the total fit from *Lawver and Scotese* [1987] (Table 5). The main point that we would like to emphasize is that the Late Cretaceous reference pole for SA is significantly different from the geographic axis and is in good agreement with results from Late Cretaceous intrusions from Brazil [*Montes-Lauar and Pacca*, 1989] and volcanics from Patagonia [*Butler et al.*, 1991]. The calculated pole at 130 Ma is slightly different from the one obtained from numerous data of the Parana flood basalts (130 Ma) [*Ernesto and Pacca.*; 1988]. The mean virtual geomagnetic pole (VGP) for the Parana Flood basalts constitutes a very well defined Early Cretaceous reference pole for South America (84.7°N, 278.5°E, $\alpha 95$=1.3; [*Ernesto and Pacca*; 1988]). It is worth noting that the mean Cretaceous pole determined by *Beck* [1988] is in fact very close to the Early Cretaceous pole from Brazil. For Jurassic time, the trend of the calculated SA APWP curve indicates a rapid northward displacement from Middle Jurassic to early Cretaceous time.

TABLE 5. Calculated Apparent Polar Wander Path for South America

Age,	VGP/Afr		rotation Afr/SA			VGP SA		
Ma	lat	long	lat	long	Angle	lat	long	α_{95}
10	84.3	159.0	59.7	-38.5	3.6	84.5	137.4	2.3
30	82.0	155.2	57.0	-35.0	11.0	80.5	107.3	2.7
50	76.9	191.1	59.0	-31.5	19.5	78.9	127.0	4.3
60	73.1	216.1	61.5	-32.5	22.7	80.5	160.7	4.2
80	66.4	232.2	63.0	-34.0	31.0	78.9	183.0	3.5
100	61.6	245.6	53.7	-33.1	41.5	84.4	174.9	7.1
130	51.0	262.4	44.5	-32.2	58.2	87.4	344.9	4.9
150	54.8	245.9	44.5	-32.2	58.2	80.3	87.8	6.4
170	63.5	231.8	44.5	-32.2	58.2	69.6	72.5	2.9
200	66.3	242.9	44.5	-32.2	58.2	71.1	56.8	6.7

APWP of Africa is from the master curve from *Besse and Courtillot* [1991]. The rotation parameters for the closure of the south Atlantic are from *Cande et al.* [1988] and the total fit from *Lawver and Scotese* [1987]; α_{95} is confidence angle from the Besse and Courtillot master curve; lat and long are latitude and longitude in degree.

Discussion

Kono et al. [1985] argue that the available paleomagnetic data define a pattern of counterclockwise rotation in Peru and

northern Chile and clockwise rotation in central and southern Chile which supports the view of an oroclinal bending of the Andean Chain. *Isacks* [1988] indicates that the large high central Andean plateau was formed by a large amount of crustal shortening. This differential shortening along the chain would result in various amounts of rotation north and south of the Bolivian deflection. However, the amount of involved rotations is not large (10° to 20°), and a best estimate of the motion of the South American plate should be taken into account to interpret paleomagnetic data from the Andes. For example, a paleomagnetic study of Cretaceous tuffs in Chile [*Beck et al.*, 1986] clearly gives results with approximately the same trend through time from 65 Ma (Chacabuco formation) to 100 Ma (San Fernando Tuff) as the one indicated by the calculated APWP. We need very precise dating of the geological units and of the acquisition of the magnetization to permit a good comparison to be made with the SA APWP because the absolute motion of the South American plate during the Cretaceous may also explain counterclockwise rotations of about 10° to 15° in the Andes.

Hereafter is a discussion and selection of reliable paleomagnetic data from the Andes of Chile and of central and southern Peru.

Data From Southern Chile

We believe, in agreement with *Forsythe et al.* [1987], that the present available data do not provide evidence for an homogeneous global clockwise rotation in Chile south of 30°S. Also, data from *Irwin et al.* [1987] and *Beck et al.* [1986] support that view. Concordant late Paleozoic magnetizations have also been reported [*Jesinkey et al.*, 1987; *Beck et al.*, 1991], casting doubt on the hypothesis of an oroclinal bending of the Chilean Andes south of 30°S.

Data From Central and Northern Chile

The most convincing data supporting clockwise rotations in northern central Chile are those of *Turner et al.* [1984] on a Cretaceous molasse sequence and data from lower Cretaceous formations [*Palmer et al.*, 1980*a*]. *Tanaka et al.* [1988] reported a few paleomagnetic results from northern Chile that support those previously published by *Turner et al.* [1984].

In northern Chile, counterclockwise rotations were first reported by *Palmer et al.* [1980*b*] on Jurassic volcanics from the Camaraca formation near Arica. Although they indicated that there is a burial metamorphism approaching the greenschist facies in the southern part of the studied area, they did not consider that the magnetization might be secondary, at least for sites showing the greatest degree of metamorphism. This is an important consideration since the applied tilt correction enhanced the deviation of the declination by about 12°. Results from the same area on dike swarms have also been reported by *Heki et al.* [1983*b*, *c*]. Remanent magnetizations from the Cuya dike swarm are not stable, and results from the Arica dike swarm suggest very little deviation of the declination. However, the same uncertainties about the primary nature of the magnetization as well as the poorly defined age of the dike swarms prevent the accurate determination of the true amount of rotation. We have shown that there is a clear remagnetization at La Yarada, and we speculate that it might also exist farther south in northern Chile.

Results From Peru

In northern Peru, results from *Heki et al.* [1984] and *Mitouard et al.* [1990] support a counterclockwise rotation of 10° to 20°.

In central Peru near Lima, a remagnetization of Late Cretaceous age was found for limestone from the Puente Piedra Formation [*May and Butler*, 1985]. The comparison of our data with the remagnetization recorded by the Puente Piedra Formation near Lima [*May and Butler*, 1985] indicates a close agreement of paleolatitude a few degrees south of the present one. However, the deviation of the declination is about 15° larger in southern coastal Peru than the one reported for the Lima area, and this would suggest a larger deformation in southern Peru than in central Peru during the Tertiary.

Primary magnetizations from Chala, Ocoña, and Arequipa indicate larger counterclockwise rotations than those recorded by the remagnetization and thus suggest that counterclockwise rotations also occurred during the Cretaceous period. However, results from the primary magnetizations at Ilo indicate that the pattern of rotations is complex.

Most of the results which have been discussed are from Cretaceous units. Thus all the Cenozoic deformations are taken into account by these results, and only data from Tertiary units can constrain the timing of the deformation. Paleomagnetic results from late Pliocene to early Miocene in southern Peru do not show any evidence of rotation (VGP: 84.1°S, 41.3°E) [*Tsunakawa et al.*, 1987].

Results From Bolivia

Previous results from three different magnetostratigraphic studies in Bolivia were reported by *McFadden et al.* [1985] and *McFadden* [1990]. They indicate that northern Bolivia has been stable since Miocene time while a clockwise rotation is found at Quebrada Honda in southern Bolivia. The large angular departure recorded by the Permian data from northern Bolivia near Lake Titicaca is not observed in late Tertiary data and might be an indication of a late Mesozoic-early Cenozoic Andean tectonic event which produced most of the rotation. This hypothesis is supported by the smaller deviation of the declination recorded by Tertiary units in southern Peru [*Heki et al.*, 1985] than those indicated by the Mesozoic units from southern Peru.

To summarize this discussion, the rotation parameters have been determined for the most significant studies (Table 6 and Figure 15); results from the primary magnetizations from Ilo and La Yarada are unclear at this stage and are not included in this list of data. Given an estimate of the age of the geological formations, the in situ rotations were calculated from the difference between the observed declination and the expected declination based on the APWP from Table 5.

CONCLUSION

Even though our critical review pointed to the problems related to the determination of the true apparent rotations across the Andes, our new data clearly support previous results reported by *Kono et al.* [1985] (Figure 15). Two tectonic models have been proposed to explain these paleomagnetic data. The oroclinal bending was suggested by *Kono et al.* [1985] and was further supported by the model of Andean chain

TABLE 6. Mean Paleomagnetic Results and Tectonic Parameters.

Site	Site location		Age	Characteristic direction			Reference pole		$R\pm\Delta R$	$F\pm\Delta F$
	Latitude	Longitude	Ma	D	I	α_{95}	Latitude °N	Longitude °E	deg	deg
Chili (1)	29.8S	70.9W	~80	9.9	-49.2	4.1	78.9	183.0	22.5±6.0	-2.4±4.5
Chili (2)	23.7S	70.4W	~130	26.2	-34.4	7.9	84.7	278.5	27.3±7.7	0.6±6.5
Peru (3)	11.9S	77.1W	~90	343.2	-28.6	3.4	78.9	183.0	-5.5±4.2	2.8±5.5
Peru (4)	13S	74W	~5-10	165.2	31.3	5.6	84.5	137.4	-11.8±5.6	-1.2±5.3
Peru (5)	11.8S	77.2W	~100	335.8	-29.9	8.3	84.4	174.9	-18.7±9.6	4.3±11.8
Bolivia (6)	17.2S	67.6W	~25	353.4	-37.4	5.4	80.5	107.3	-7.5±6.0	-7.7±5.1
Bolivia (6)	17.5S	67W	~7	355.2	-25.8	5.9	84.5	137.4	-2.3±5.6	-13.8±5.4
Bolivia (6)	22S	65.6W	~12	17.8	-40.7	3.9	84.5	137.4	20.2±4.6	-4.9±3.9
This study										
Chala (r)	15.8S	74.3W	~80	324.9	-25.1	4.4	78.9	183.0	-23.7±4.9	-7.8±5.6
Chala (p)	15.8S	74.3W	~170	290.5	-31.3	16.0	69.6	72.5	-82.6±15.3	-20.5±13.0
Ilo (r)	17.5S	71.4W	~80	330.9	-39.9	8.5	78.9	183.0	-17.7±9.4	3.6±8.0
La Yarada (r)	18.1S	70.7W	~80	330.6	-38.1	6.0	78.9	183.0	-18.0±6.8	0.8±6.3
Aréquipa	16.5S	71.8W	~100	299.5	-47.0	15.4	84.4	174.9	-55.1±19.3	13.0±15.1

1, *Palmer et al.*, [1980*a*]; 2, *Turner et al.*,[1984] (normal and reverse combined); 3, *May and Butler*, [1985]; 4, *Heki et al.*, [1985]; 5, *Heki et al.*, [1984] and 6, *McFadden* [1990]. $R\pm\Delta R$, $F\pm\Delta F$: rotation and flattening parameters and their associated errors [*Beck*, 1980; *Demarest*, 1983]; r is remagnetization component; p is primary component; D,I are declination and inclination of the characteristic paleomagnetic direction in degree.

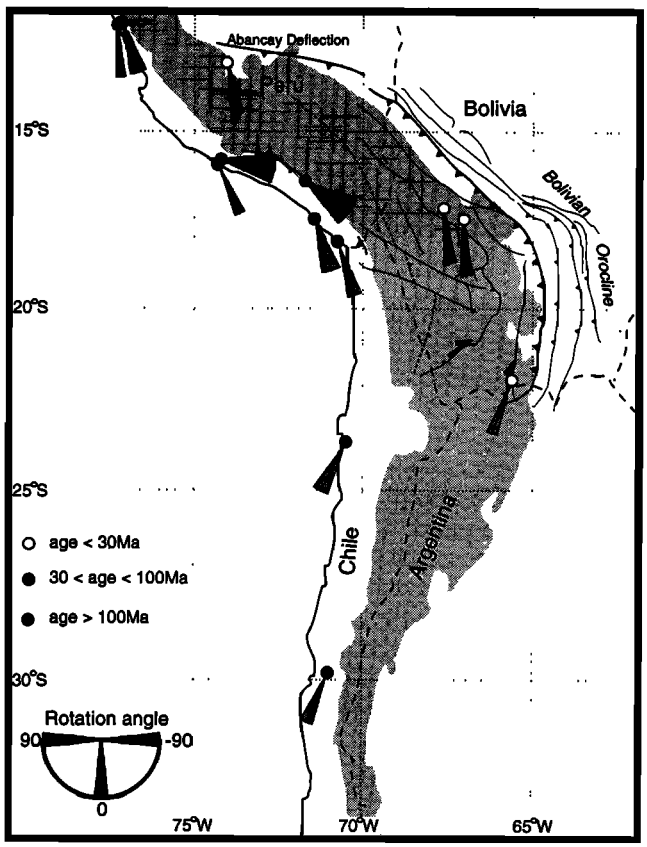


Fig. 15. Inferred apparent rotations and associated errors along the Bolivian orocline (data from Table 6). The shaded area corresponds to the area with average elevation above 3 km [from *Isacks*, 1988]. The structural fault system has been redrawn from *Sempéré et al.* [1991].

building by *Isacks* [1988]. On the contrary, *Beck* [1988] advocated for rotations in a strike-slip system induced by nonorthogonal convergence of the subducting oceanic plate.

Our results from Peru and Bolivia show evidence for a two-stage formation of the Bolivian orocline. Results from primary magnetizations in Jurassic volcanics and plutonic intrusions suggest that large rotations occurred in southern Peru and Bolivia possibly during the Late Cretaceous. *Soler et al.* [1989] speculated that the underplating of an aseismic oceanic plateau in southern Peru during the Late Cretaceous has induced the thrusting of the Arequipa massif over the continental plate. This underplating might have occurred at a time where the convergence was more oblique toward the northeast, and it might have acted as a buttress. The incipient orocline has then evolved to its present attitude by differential shortening along the Andean margin during the Cenozoic that has produced a regional counterclockwise rotation of 15-20° recorded by the secondary magnetization in southern Peru.

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